Chapter 1: Real Numbers

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Welcome to the foundation of Class 10 Mathematics! Real numbers are the ones we use in our daily lives—they include integers, fractions, rational sequences, and those mysterious irrational quantities. Let's conquer the fundamental theorems that govern them!

Master Mind Map 🗺️

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graph TD
    RN[Real Numbers] --> RA[Rational Numbers]
    RN --> IR[Irrational Numbers]

    RA --> FTA[Fundamental Theorem of Arithmetic]
    FTA --> PF[Prime Factorization]

    PF --> HCF[Highest Common Factor HCF]
    PF --> LCM[Lowest Common Multiple LCM]

    HCF --> PROP[Property: HCF × LCM = a × b]

    IR --> PI[Proofs of Irrationality]
    PI --> P2[E.g., Proof of √2 is irrational]
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1. The Fundamental Theorem of Arithmetic 🧮

Every composite number can be expressed (factorized) as a product of primes, and this factorization is **unique**, apart from the order in which the prime factors occur.

- **Composite Number:** A number that has factors other than 1 and itself (e.g., 4, 6, 8, 9).

- **Prime Number:** A number that is divisible only by 1 and itself (e.g., 2, 3, 5, 7, 11).

> [!NOTE] The number **1** is neither prime nor composite! The number **2** is the only even prime number.

Finding HCF and LCM using Prime Factorization

Once you have the prime factors of numbers, finding their HCF and LCM is a breeze.

1. **HCF (Highest Common Factor):** Product of the smallest power of each *common* prime factor in the numbers.
2. **LCM (Lowest Common Multiple):** Product of the greatest power of *each* prime factor involved in the numbers.

Example: Find HCF and LCM of 6 and 20

- $6 = 2^1 \times 3^1$
- $20 = 2^2 \times 5^1$
- **HCF:** Common factor is 2. Smallest power is $2^1 = 2$.
- **LCM:** Product of highest powers of all factors = $2^2 \times 3^1 \times 5^1 = 4 \times 3 \times 5 = 60$.

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2. The Golden Property of HCF and LCM 🥇

For any two positive integers a and b, there is a beautiful relationship between the numbers and their HCF and LCM:

$$ \text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b $$

> [!TIP] This formula is highly tested in board exams! If you are given any three of the values (e.g., the two numbers and their HCF), you can instantly find the fourth (the LCM) by dividing: $\text{LCM} = \frac{a \times b}{\text{HCF}}$.

> [!WARNING] This property ONLY holds true for exactly **two** numbers. **Do not** apply $HCF \times LCM = a \times b \times c$ for three numbers!

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3. Revisiting Irrational Numbers 🤯

A number is called **irrational** if it cannot be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

- Examples: $\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi$.

Theorem: Proof by Contradiction

Let p be a prime number. If p divides a^2, then p also divides a, where a is a positive integer. We use this theorem deeply when proving roots are irrational.

Standard Proof: Provo that $\sqrt{2}$ is irrational.

1. **Assumption:** Let us assume, to the contrary, that $\sqrt{2}$ is rational.
2. That means we can find coprime integers a and b ($b \neq 0$) such that $\sqrt{2} = \frac{a}{b}$. (Coprime means their only common factor is 1).
3. Squaring both sides: $2 = \frac{a^2}{b^2} \implies 2b^2 = a^2$.
4. This means 2 divides a^2, so **2 divides a**.
5. So, we can write $a = 2c$ for some integer c.
6. Substitute this back: $2b^2 = (2c)^2 \implies 2b^2 = 4c^2 \implies b^2 = 2c^2$.
7. This means 2 divides b^2, so **2 divides b**.
8. **Contradiction:** But a and b have at least 2 as a common factor! This contradicts our assumption that they were coprime.
9. Therefore, our assumption was wrong, and $\sqrt{2}$ is **irrational**.

Quick Revision Notes 📌

1. **Prime factorization** is unique for every composite number.
2. **HCF:** Lowest power of *common* primes.
3. **LCM:** Highest power of *all* primes.

4. **Verification Formula:** $\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$.
5. **Irrationality proofs** primarily rely on assuming the number is rational ($\frac{a}{b}$) and forcing a contradiction regarding integer division and coprimes.

Chapter Summary 🏁

Real numbers divide cleanly into rational and irrational categories. For rational numbers, the Fundamental Theorem of Arithmetic ensures that every composite number breaks down uniquely into prime factors, providing an elegant way to identify HCFs and LCMs. We can verify our calculations easily using the product property. For irrational numbers, we utilize logical proofs by contradiction to definitively establish that numbers like $\sqrt{2}$ and $\sqrt{3}$ can never be expressed as simple fractions. Mastery of these structural proofs forms the logical foundation for advanced mathematics!